

Square Status Sync Webhooks Sprint

Date: 2026-07-09

Status: Complete

Summary

Implemented one-way Square → Open Order fulfillment status sync using Square `order.updated` webhooks. Status changes made in Square Order Manager, POS, or Dashboard update the matching `VendorOrder` in Open Order. Checkout, payouts, Square order injection, Deliverect routing, and manual/tablet routing are unchanged.

Files changed

Area	Files
Webhook route	<code>src/app/api/webhooks/square/route.ts</code> , <code>square-webhook-route.test.ts</code>
Signature verify	<code>src/lib/integrations/square/square-webhook-verify.ts</code> , <code>.test.ts</code>
Payload parsing	<code>src/lib/integrations/square/square-webhook-payload.ts</code>
Status mapper	<code>src/lib/integrations/square/square-status-mapper.ts</code> , <code>.test.ts</code>
Sync service	<code>src/services/square-status-sync.service.ts</code> , <code>.test.ts</code>
Square API	<code>src/lib/integrations/square/square-api.client.ts</code> (<code>fetchSquareOrder</code>)
Types / audit	<code>src/lib/integrations/square/square-order.types.ts</code> , <code>square-order-audit.ts</code>
Schema	<code>prisma/schema.prisma</code> , migration <code>20260709120000_square_webhook_status_source</code>
Status instrumentation	<code>src/services/vendor-order-status-instrumentation.ts</code> , <code>src/domain/status-authority.ts</code>
Admin UI	<code>AdminDeliverectDiagnosticsPanel.tsx</code> , <code>AdminSquareStatusSync.tsx</code>
Admin manual sync	<code>src/app/api/admin/vendor-orders/[vendorOrderId]/square-status-sync/route.ts</code>
Vendor UI	<code>VendorOrderDetailPanel.tsx</code> , <code>vendor-orders-board-data.ts</code> , <code>deliverect-vendor-order-authority.ts</code>
Env	<code>src/lib/env.ts</code> (<code>SQUARE_WEBHOOK_NOTIFICATION_URL</code>)

Docs	docs/integrations/square-status-sync.md , square-production-checklist.md , docs/ops/order-routing-env-vars.md
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Webhook endpoint

POST /api/webhooks/square

- Verifies `x-square-hmacsha256-signature` using `SQUARE_WEBHOOK_SIGNATURE_KEY` + `SQUARE_WEBHOOK_NOTIFICATION_URL`
- Rejects invalid signatures (`403` , logs `failed_signature`)
- Returns `503` when webhook env not configured (does not affect checkout or other routing)
- Primary event: `order.updated`
- Ignores duplicate `event_id` via `ProviderWebhookEvent` unique constraint

Event persistence / idempotency

Uses `ProviderWebhookEvent` (`provider=square`):

Field	Purpose
<code>externalEventId</code>	Square <code>event_id</code> — duplicate-safe
<code>externalObjectId</code>	Square order id
<code>eventType</code>	e.g. <code>order.updated</code>
<code>processingStatus</code>	<code>received</code> / <code>processed</code> / <code>ignored</code> / <code>failed</code>
<code>errorCode</code>	<code>ignored_no_match</code> , <code>ignored_non_order_update</code> , <code>failed_signature</code> , <code>failed_processing</code> , etc.
<code>relatedVendorOrderId</code>	Linked after match

Vendor order audit: `lastSquarePayload.statusSync`
(`SquareWebhookLastApplyRecord`).

Status mapping

Square pickup fulfillment	OO fulfillment	OO routing (when progressing)
PROPOSED	accepted	confirmed
RESERVED	preparing	confirmed
PREPARED	ready	confirmed
COMPLETED	completed	confirmed
CANCELED / FAILED	cancelled	confirmed

Monotonic merge: terminal OO `completed` / `cancelled` do not regress. Fetches latest order via `GET /v2/orders/{id}` before applying.

OO status update path

1. Webhook or admin manual sync
 2. `syncSquareOrderStatusBySquareOrderId` / `applySquareOrderStatusSync`
 3. Vendor OAuth `fetchSquareOrder`
 4. `mapSquareOrderSnapshotToVendorStatus` + `mergeSquareMappedIntoVendorOrder`
 5. `applyVendorOrderStatusWithMeta` (`statusSource: square_webhook` , `historySource: square`)
 6. `evaluateCustomerOrderMilestones` → preparing/ready SMS (existing idempotency keys)
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SMS / idempotency

Square updates flow through the same `applyVendorOrderStatusWithMeta` → `evaluateCustomerOrderMilestones` path as vendor dashboard actions. Duplicate webhooks produce `noop_same_status` and do not re-send SMS.

Admin observability

Admin order detail → Square routing details:

- Status sync configured (yes/no)

- Last synced at, fulfillment/order state, last error
 - **Sync status from Square now** button
 - Collapsible payload/response audit (unchanged)
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Vendor dashboard

Square-routed orders with `squareOrderId` show:

"This order is routed to Square. Status updates from Square will update Open Order."

OO kitchen action buttons remain available (MVP — no hard lock). Deliverect POS-lock copy is not shown for Square vendors.

Tests / QA

#	Scenario	Result
1	Signature verification passes	✓
2	Invalid signature rejected	✓
3	Duplicate event ID ignored	✓
4	No matching <code>squareOrderId</code> → ignored	✓
5	Matching order fetches Square order	✓
6	PREPARED → OO ready	✓
7	COMPLETED → OO completed	✓
8	Cancellation maps to cancelled	✓
9	Terminal statuses do not regress	✓
10	Admin manual sync uses same mapper	✓
11	Missing signature key → 503, no checkout impact	✓
12	Deliverect/manual routing tests unchanged	✓
13	<code>npm run build</code>	✓

31 new/related unit tests passing in Square status sync suite.

Known limitations

1. One-way sync only — OO status edits do not push to Square
 2. Only `order.updated` in this sprint (not `order.fulfillment.updated` or `payment.updated`)
 3. Webhook URL must match Square subscription exactly for signature verification
 4. Requires prior Square injection (`squareOrderId` on `VendorOrder`)
 5. Vendor kitchen buttons not disabled — staff can still update in OO (possible dual-system edits)
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Rollout

1. `npx prisma migrate deploy`
2. Square Developer Dashboard → webhook subscription → `order.updated` → `https://www.openorderco.com/api/webhooks/square`
3. Set `SQUARE_WEBHOOK_SIGNATURE_KEY` and `SQUARE_WEBHOOK_NOTIFICATION_URL` in production
4. Sandbox QA per `docs/integrations/square-status-sync.md`